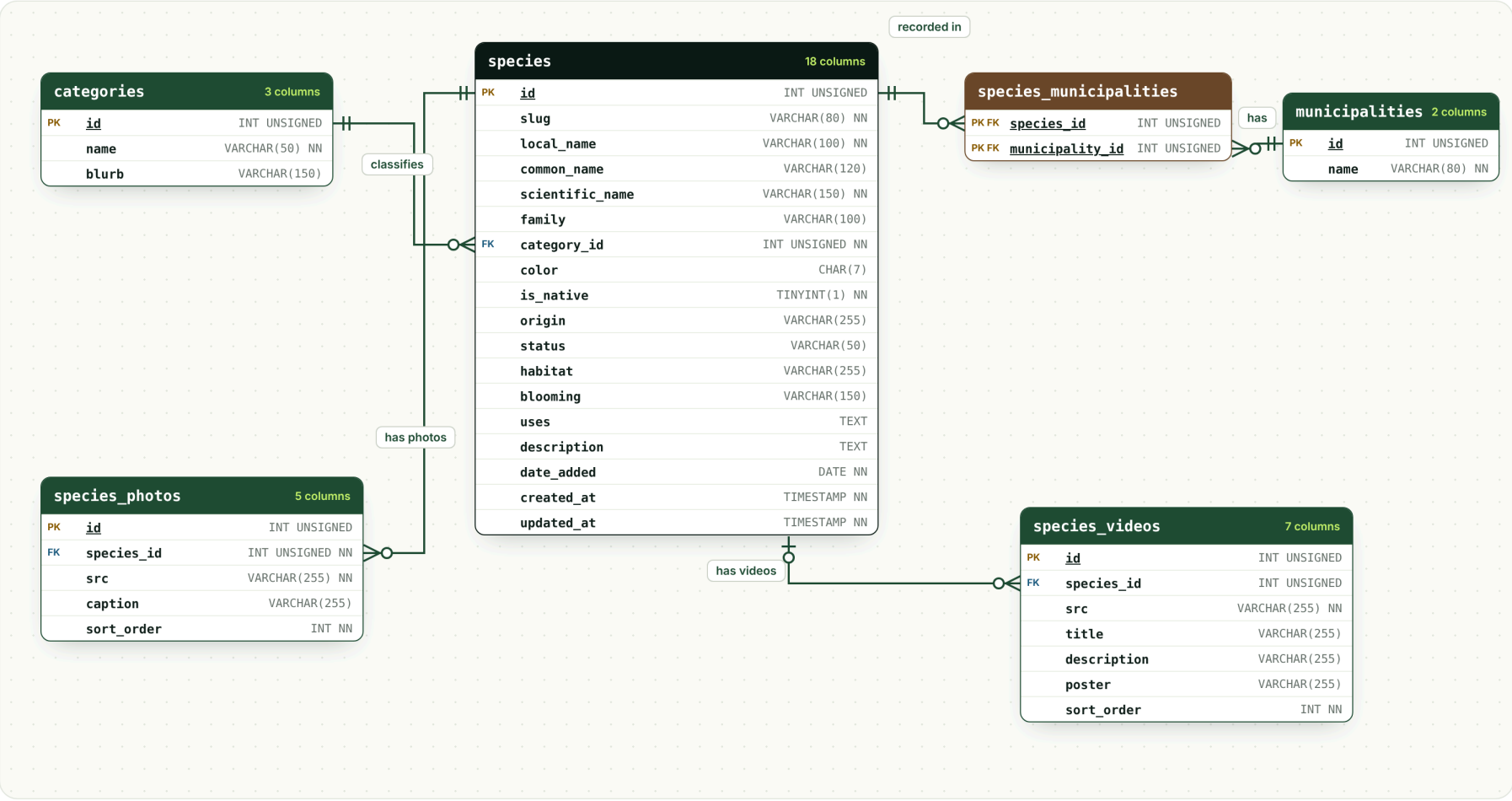
Siargao Flora — Database Design

System Integration and Architecture · Final Project · Database **siargao_flora** (MySQL, InnoDB, utf8mb4) · SQL file: `database/siargao_flora.sql`

Entity Relationship Diagram

Database **siargao_flora** · MySQL · 6 tables · 5 relationships

Siargao Flora



PK Primary key FK Foreign key underlined = primary key column exactly one zero or one zero or many

Data dictionary

Every table and column with its data type, key and description. NN = NOT NULL, PK = primary key, FK = foreign key.

`categories` Plant groups · 5 records

COLUMN	DATA TYPE	NULL	KEY	DEFAULT / EXTRA	DESCRIPTION
`id`	`INT UNSIGNED`	NO	PK	`auto_increment`	Primary key
`name`	`VARCHAR(50)`	NO	UNIQUE		Group name, e.g. Ornamental
`blurb`	`VARCHAR(150)`	YES			Short description of the group

`municipalities` Municipalities of Siargao · 9 records

COLUMN	DATA TYPE	NULL	KEY	DEFAULT / EXTRA	DESCRIPTION
`id`	`INT UNSIGNED`	NO	PK	`auto_increment`	Primary key
`name`	`VARCHAR(80)`	NO	UNIQUE		Municipality name, e.g. General Luna

species Plants and flowers recorded on Siargao Island · 20 records

COLUMN	DATA TYPE	NULL	KEY	DEFAULT / EXTRA	DESCRIPTION
id	INT UNSIGNED	NO	PK	auto_increment	Primary key
slug	VARCHAR(80)	NO	UNIQUE		URL identifier, e.g. gumamela (species.php?id=gumamela)
local_name	VARCHAR(100)	NO			Local (Bisaya/Filipino) name
common_name	VARCHAR(120)	YES			English common name
scientific_name	VARCHAR(150)	NO			Scientific (Latin) name
family	VARCHAR(100)	YES			Botanical family
category_id	INT UNSIGNED	NO	FK → categories.id		FK → categories.id (plant group)
color	CHAR(7)	YES		#5b8a4a	Hex colour used for the website illustration
is_native	TINYINT(1)	NO		0	1 = native to Asia, 0 = introduced
origin	VARCHAR(255)	YES			Where the plant originally comes from
status	VARCHAR(50)	YES			Optional status, e.g. Endangered
habitat	VARCHAR(255)	YES			Where it grows
blooming	VARCHAR(150)	YES			Blooming / fruiting season
uses	TEXT	YES			Uses and significance
description	TEXT	YES			Description of the plant
date_added	DATE	NO		CURRENT_DATE	Date the record was added
created_at	TIMESTAMP	NO		CURRENT_TIMESTAMP	Row created
updated_at	TIMESTAMP	NO		CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Row last changed

species_municipalities Link table: municipalities where each species was recorded · 20 records

COLUMN	DATA TYPE	NULL	KEY	DEFAULT / EXTRA	DESCRIPTION
species_id	INT UNSIGNED	NO	PK, FK → species.id		PK + FK → species.id
municipality_id	INT UNSIGNED	NO	PK, FK → municipalities.id		PK + FK → municipalities.id

species_photos Species photos · 20 records

COLUMN	DATA TYPE	NULL	KEY	DEFAULT / EXTRA	DESCRIPTION
id	INT UNSIGNED	NO	PK	auto_increment	Primary key
species_id	INT UNSIGNED	NO	FK → species.id		FK → species.id
src	VARCHAR(255)	NO			File name in assets/images/species/ or a full URL
caption	VARCHAR(255)	YES			Caption, usually where the photo was taken
sort_order	INT	NO		0	Display order (0 = main photo)

species_videos Field videos · 3 records

COLUMN	DATA TYPE	NULL	KEY	DEFAULT / EXTRA	DESCRIPTION
id	INT UNSIGNED	NO	PK	auto_increment	Primary key
species_id	INT UNSIGNED	YES	FK → species.id		FK → species.id (NULL = general field footage)
src	VARCHAR(255)	NO			File name in assets/videos/, a full URL or a YouTube link
title	VARCHAR(255)	YES			Video title
description	VARCHAR(255)	YES			What the video shows
poster	VARCHAR(255)	YES			Cover image file name in assets/videos/
sort_order	INT	NO		0	Display order

Relationships

RELATIONSHIP	TYPE	FOREIGN KEY	ON DELETE	MEANING
`categories → species`	One-to-many	`species.category_id → categories.id`	RESTRICT	Each species belongs to exactly one plant group; a group has zero or many species.
`species ↔ municipalities`	Many-to-many	`species_municipalities.species_id → species.id` `species_municipalities.municipality_id → municipalities.id`	CASCADE	A species can be recorded in many municipalities, and a municipality can have many species. Resolved with the link table `species_municipalities` (composite primary key).
`species → species_photos`	One-to-many	`species_photos.species_id → species.id`	CASCADE	A species has zero or many photos; each photo belongs to one species.
`species → species_videos`	One-to-many (optional)	`species_videos.species_id → species.id` (nullable)	CASCADE	A species has zero or many videos. A video with no species (NULL) is general field footage.

Collected records

20	5	9	20	20	3
species	plant groups	municipalities	species–town links	photos	videos

#	LOCAL NAME	COMMON NAME	SCIENTIFIC NAME	FAMILY	PLANT GROUP	RECORDED IN
1	**Baging-baging**	String of Nickels	*Dischidia nummularia*	Apocynaceae	Ornamental	San Isidro
2	**Bayabas**	Guava	*Psidium guajava*	Myrtaceae	Fruit	Del Carmen
3	**Bromeliad**	Blushing Bromeliad	*Neoregelia carolinae*	Bromeliaceae	Ornamental	General Luna
4	**Bugambilya**	Paper Flower (Bougainvillea)	*Bougainvillea glabra*	Nyctaginaceae	Ornamental	General Luna
5	**Dragon Fruit**	Dragon Fruit (Pitaya)	*Selenicereus undatus*	Cactaceae	Fruit	Del Carmen
6	**Gumamela**	Double-Flowered Chinese Hibiscus	*Hibiscus rosa-sinensis*	Malvaceae	Ornamental	General Luna
7	**Kamatis**	Tomato	*Solanum lycopersicum*	Solanaceae	Vegetable	Dapa
8	**Kapayas**	Papaya	*Carica papaya*	Caricaceae	Fruit	Pilar
9	**Kawayan**	Bamboo	*Bambusa vulgaris*	Poaceae	Ornamental	Pilar
10	**Maguey**	Mauritius Hemp	*Furcraea foetida*	Asparagaceae	Ornamental	Del Carmen
11	**Niyog**	Coconut	*Cocos nucifera*	Arecaceae	Palm	General Luna
12	**Orkid**	Orchids	*Orchidaceae*	Orchidaceae	Ornamental	Dapa
13	**Pako**	Bird's Nest Fern	*Asplenium nidus*	Aspleniaceae	Ornamental	General Luna
14	**Sahod-Yaman**	Tree Philodendron	*Thaumatophyllum bipinnatifidum*	Araceae	Ornamental	Pilar
15	**Sawa-Sawa**	Snake Plant	*Dracaena trifasciata*	Asparagaceae	Ornamental	General Luna
16	**Sili**	Chili Pepper	*Capsicum frutescens*	Solanaceae	Vegetable	General Luna
17	**Talong**	Eggplant	*Solanum melongena*	Solanaceae	Vegetable	Dapa
18	**Tangkong**	Water Spinach	*Ipomoea aquatica*	Convolvulaceae	Vegetable	Pilar
19	**Tanglad**	Lemongrass	*Cymbopogon citratus*	Poaceae	Herb	Pilar
20	**Umbrella Plant**	Variegated Dwarf Umbrella Plant	*Heptapleurum arboricola*	Araliaceae	Ornamental	General Luna